

All About The Back



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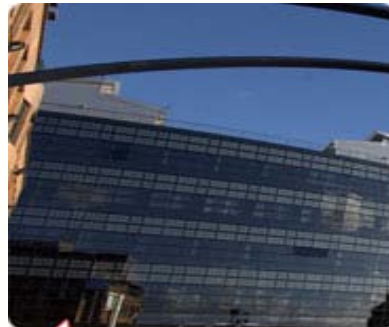
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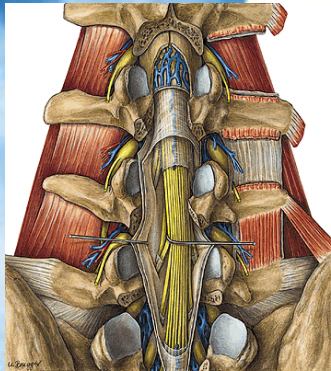
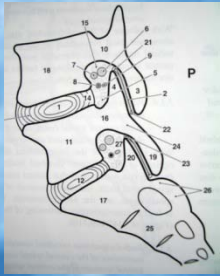
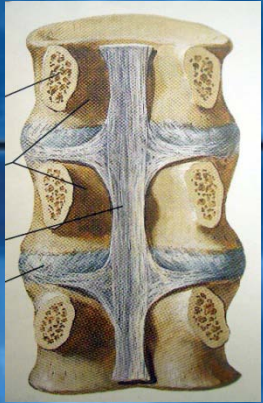
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Spine: Biomechanics



- **Osteo-Liagamentous Complex (Bones&Co)**
 - Crucial for Stabillity
- **FSU (Functional Spinal Unit) 3-Joint Complex:**
 - Discs: Load-bearing (core)+Tensile loads(annulus)
 - 2 Joints: Posterior load path, Compressive Load 10-20%
 - Buckling CSpine 10N / TSpine 20N/ LSpine 88N
- **Muscles:** Superficial / Deep Layer
- **Posture:** CSpine and LSpine Lordosis
 - supine: 200N
 - Standing with 30grd flexion: 1400N

Sources of spinal pain

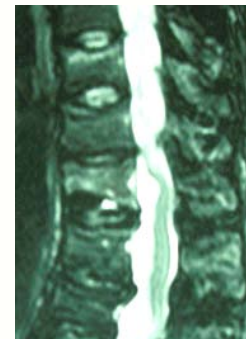
Neoplastic: neuroma, meningioma



Vascular
Paraspinal
Pathology



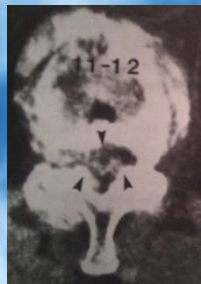
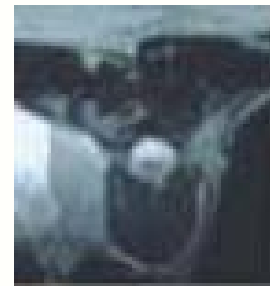
Osteoporosis



Infectious



Traumatic



Degenerative



Spine: Sources of Pain in Degenerative Conditions

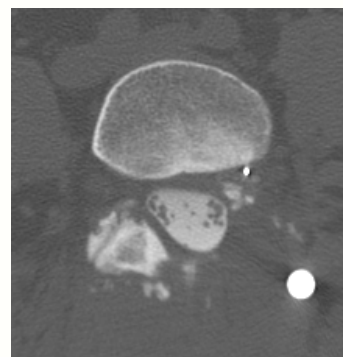
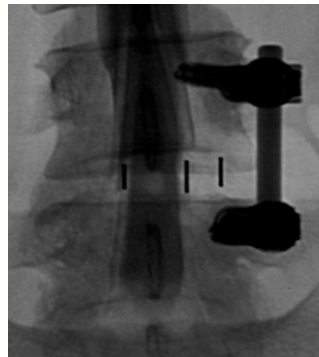
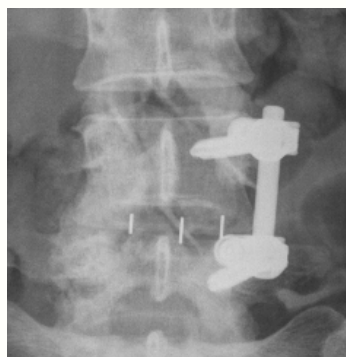
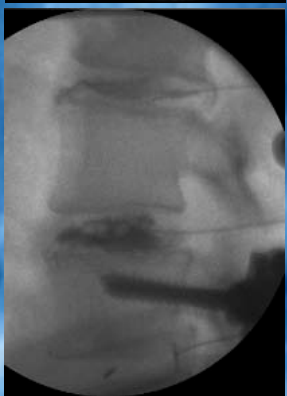
- **Cause of Pain**
 - 19th and mid 20th century: many different aetiologies
 - Since 1950: disc degeneration main driver
- Tendency to prefer **organic reasons**
- Inclination to trust **technical diagnostic** results over clinical judgement
- But: obvious causality and predisposition between **psychiatric disorders** and chronic back pain

The Mind or The Spine-Which goes first?

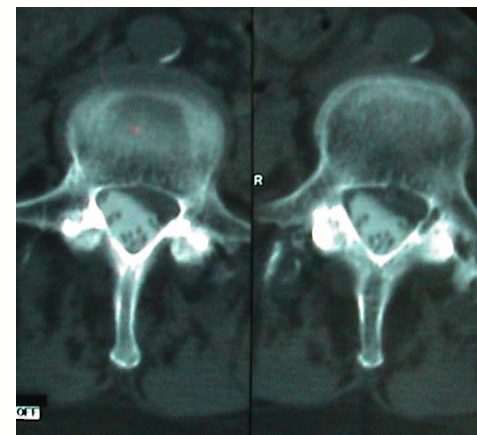
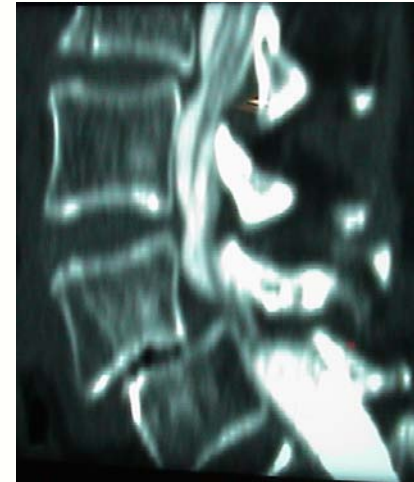
- 200 patients
- current and lifetime psychiatric syndromes
 - 77% lifetime criteria
 - 59% current symptoms
 - Major depression: 51% before pain onset
 - Substance abuse: 94%
 - Anxiety disorder: 95%

Spine: Establishing the Objective Facts

- X Rays + dynamic views (load-bearing!)
- CT +/- Myelography
- MRI +/- Gad + dynamic views
- Provocative Studies
- Diagnostic Blocks
- NCS / EMG



XRay-Myelography and CT-Myelography



Spine: Establishing the Objective Facts

Dynamic MRI

Sit Neutral



Sit Flexion



Sit Extension



BUT: **MRI in ASYMPTOMATIC Subjects**

**Findings in group of middle-aged people
(45 yo average age):**

Disc Pathology	57-64%
• Disc bulges	39-53%
• Disc protrusion	18-63%
• Disc extrusion	13-18%
• Disc sequestration	0%
• Nerve root compression	4%
DDD (at least one level)	85%

Boos et al, Jensen et al, Wood et al

Spine Surgery: Epidemiology & AEtiology

- Lumbar spine interventions /year (US)
 - >250,000 for persistent sciatica
 - >40,000 catheter procedures discogenic pain
 - >300,000 for back pain incldg.
 - >190,000 fusions
- Cost of treatment >170 billions of US\$

Dewing et al, Spine. 2008;33(1):33-38

Leigh et al., Arch Intern Med 1997;157-68

Freeman et al., Spine. 2005;30(21):2369-2377 National Hospital Survey, 1997. Series 13, No. 144

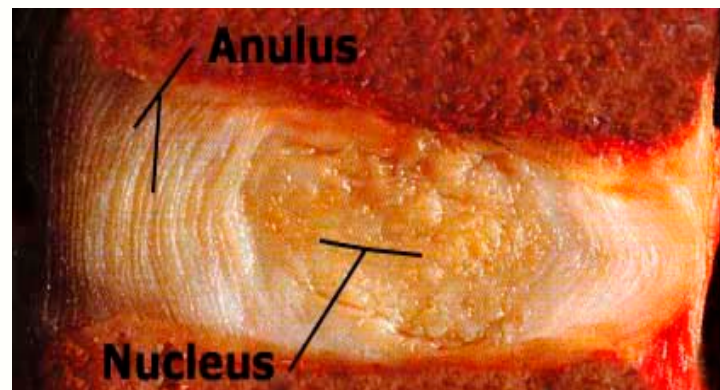


Degenerative Disorders of the Adult Spine

- **Cervical Degenerative Disorders**
 - Cervical Disc Herniation and Stenosis
 - Cervical Myelopathy
- **Thoracic Degenerative Disorders**
 - Thoracic Disc Herniation and Stenosis
- **Lumbar Degenerative Disorders**
 - Low Back Pain
 - Discogenic Lumbar Pain Syndrome
 - Lumbar Disc Herniation and Stenosis

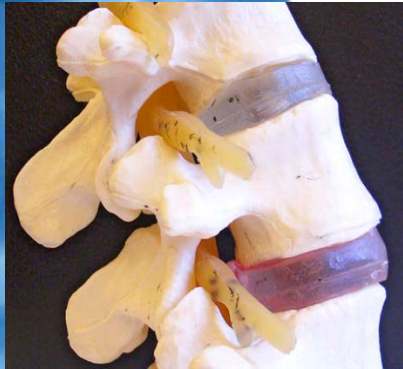
Highest Vulnerability

- Age 40 –49
- Level L4/5 or L5/S1 (75%)
- Movement pattern Flexion +Rotation
- Degree of DDD Stage 2 –Laxity

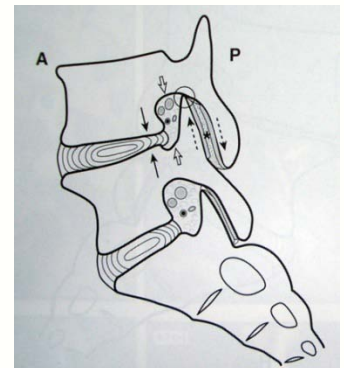
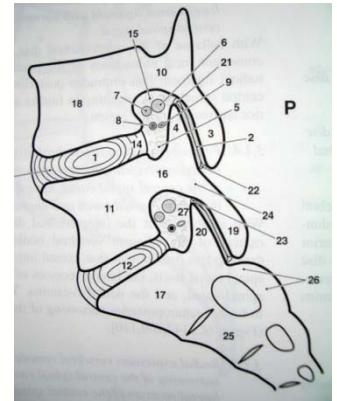


Kirkaldy-Willis (1983) Managing LBP

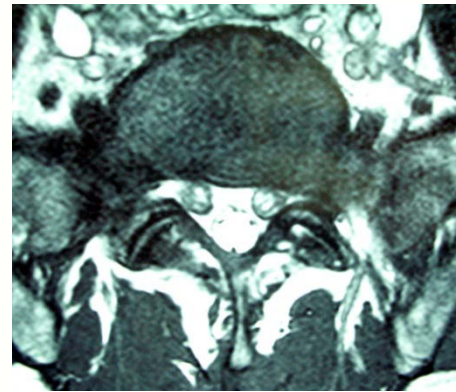
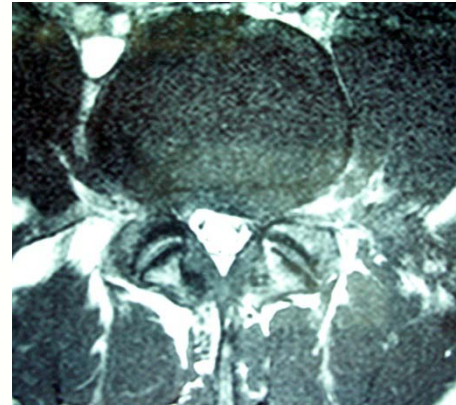
Common Causes of chronic LBP associated with degenerative disc disease (DDD)



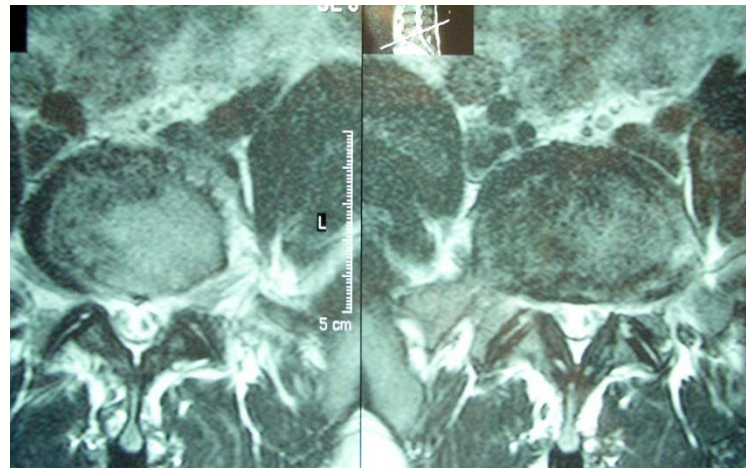
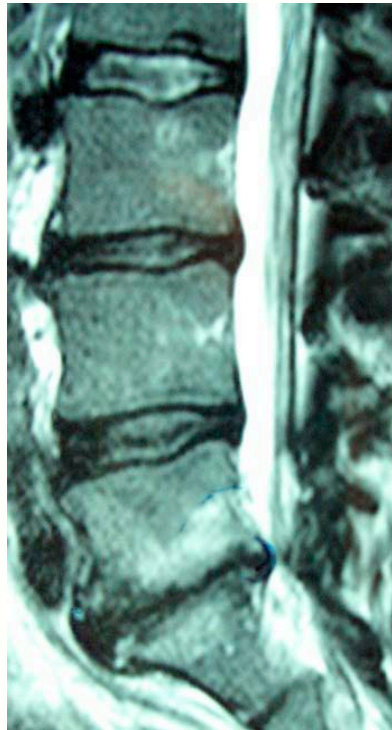
- Intervertebral Disc
- Facet Joints
- Nerve Roots
- Ligaments, Muscles
- SI-Joint



MR Imaging and DDD (moderate)



MR Imaging and DDD (severe)



Clinical Sx of patients with DDD

- Load-dependent LBP
- $\pm$ radicular pain
- Pain exacerbated by changing body positions, F/E, rotation, axial load
- Paravertebral spasm most cases
- Anterior tenderness of affected level
- Focal deficits are rare
- ROM is decreased with F/E+lateral bending

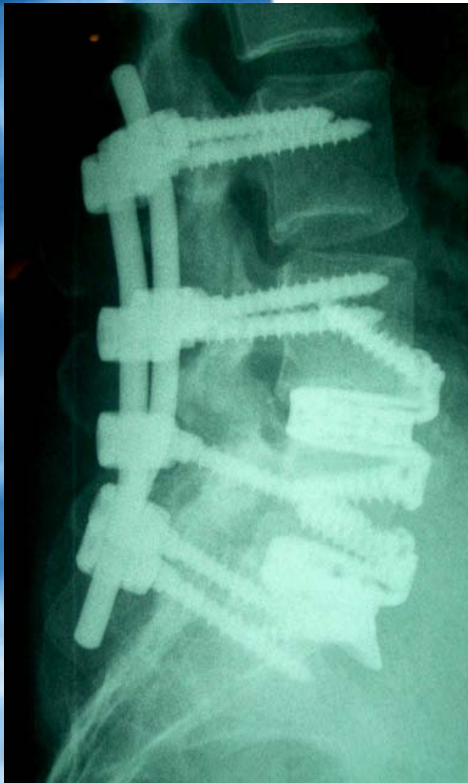
F/E= Flexion / Extension

Differentiating Pain as a Sx

- Radicular Pain (extradural pain)
 - Unilateral
 - Dermatomal in distribution
 - Lancinating, shooting, sharp pain
- Central Pain (intramedullary pain)
 - Deep, poorly defined
 - Intractable
 - No dermatomal relation

Elimination of Pain Source can be achieved by:

- Conservative Tx
- Removal of Nociceptor
 - (discectomy, facetectomy, spinal canal decompression)
- Fixation
 - e.g. segmental fusion
- Remodeling
 - Reduction with fusion



Surgical Management of the Spine

Indications for Intervention

- Cauda Equina Syndrome (CES) 1-2% of all pts.
retention/incontinence/
saddle anaesthesia/erectile d/o
outcome: only 50% regain function
- Progressive motor deficit
- Intractable pain
- Instability
- Revision procedure

Annular Tears and Surgical Options

